

Maximum Likelihood Line Spectral Estimation in the Signal Domain: A Rank-Constrained Structured Matrix Recovery Approach

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Abstract—Maximum likelihood estimation (MLE) provides a well-known benchmark for line spectral estimation and has been extensively studied in the parameter domain using a variety of optimization algorithms. To overcome the sensitivity of these algorithms to parameter initialization, in this paper we study the MLE in the signal domain. We formulate the MLE as an equivalent rank-constrained structured matrix recovery problem that admits a unique matrix solution containing the signal, from which the parameters of interest are uniquely retrieved. The alternating direction method of multipliers (ADMM) is used to solve the rank-constrained problem and it is shown to have a good convergence behavior. The proposed approach is generalized to the case of missing data and arbitrary-dimensional line spectral estimation. Extensive numerical results are provided that corroborate our analysis and confirm that the proposed approach globally solves the MLE problem and outperforms state-of-the-art algorithms.

Index Terms—Line spectral estimation, maximum likelihood estimation, Hankel-Toeplitz optimization model, rank-constrained structured matrix recovery, ADMM, nonconvex optimization.

I. INTRODUCTION

LINE spectral estimation is a fundamental problem in statistical signal processing [1]. It aims at estimating the frequencies and coefficients of complex sinusoids from regularly spaced noisy samples of their superposition. The noisy samples $\{y_j\}$ are given by

$$y_j = \sum_{k=1}^K s_k e^{i2\pi f_k(j-1)} + \varepsilon_j, \quad j = 1, 2, \dots, N, \quad (1)$$

where $i = \sqrt{-1}$, $\varepsilon_j \in \mathbb{C}$ is additive measurement noise, $f_k \in [0, 1)$ and $s_k \in \mathbb{C}$ denote the frequency and coefficient of the k -th spectral component, respectively. Our objective is to estimate $\{f_k\}$ and $\{s_k\}$ given $\{y_j\}$ (or a subset of $\{y_j\}$) and

the model order K . This problem is also known as spectral super-resolution [2] or spectral compressive sensing [3], [4] and together with its multidimensional counterparts is encountered in a variety of applications, including radar [5], [6], sonar [7], wireless communications [8], and channel sounding [9].

The research on line spectral estimation has a long history; see [1]. Within the statistical framework for parameter estimation, the maximum likelihood estimation (MLE) is asymptotically efficient and serves as a fundamental benchmark for line spectral estimation [1], [10]. By assuming deterministic (but unknown) frequencies $\{f_k\}$ and coefficients $\{s_k\}$ and random additive white Gaussian noise $\{\varepsilon_j\}$, the (deterministic/conditional) MLE can be formulated as the following nonlinear least squares (NLS) problem [11]:

$$\begin{aligned} \min g(f_k, s_k) &= \sum_{j=1}^N \left| y_j - \sum_{k=1}^K s_k e^{i2\pi f_k(j-1)} \right|^2 \\ &= \left\| \mathbf{y} - \sum_{k=1}^K \mathbf{a}(f_k) s_k \right\|_2^2, \end{aligned} \quad (2)$$

where $\mathbf{a}(f_k) = [1, e^{i2\pi f_k}, \dots, e^{i2\pi f_k(N-1)}]^T \in \mathbb{C}^N$ and $\mathbf{y} \in \mathbb{C}^N$ is a vector defined by stacking all y_j 's. Since $\{s_k\}$ can easily be estimated by a least squares method given $\{f_k\}$, the key is to solve for $\{f_k\}$. This is difficult since the objective function in (2) is highly nonconvex with respect to $\{f_k\}$ and has a *complicated multimodal shape with a very sharp global minimum* [12]. We also mention that solving the MLE lies at the core of information-theoretic criteria for determining the model order K if it is unknown [13], which is another fundamental problem in line spectral estimation.

Several optimization methods have been proposed to solve (2), including the Newton's method (NM) [14], expectation maximization (EM) [15], alternating projection (AP) [16], the RELAX algorithm [17], newtonized orthogonal matching pursuit (NOMP) [18] and so on. We call these methods parameter-domain approaches since they solve directly for the optimal frequencies. Among them, NM, EM and AP are sensitive to parameter initialization, which limits their practical interest [1]. RELAX and NOMP can be seen as iterative greedy algorithms with a feedback mechanism and they do not require an explicit initialization [18].

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Differently from the aforementioned parameter-domain approaches and inspired by the recent gridless sparse methods [19], in this paper we formulate the MLE as the following signal recovery problem under a spectral sparsity constraint:

$$\min_{\mathbf{x} \in \mathbb{C}^N} \|\mathbf{y} - \mathbf{x}\|_2^2, \text{ subject to } \mathbf{x} \in S_0, \quad (3)$$

where

$$S_0 = \left\{ \sum_{k=1}^K \mathbf{a}(f_k) s_k : f_k \in [0, 1), s_k \in \mathbb{C} \right\}. \quad (4)$$

To overcome the high nonlinearity with respect to $\{f_k\}$ and make (3) a tractable optimization problem, the key is to formulate the constraint $\mathbf{x} \in S_0$ in (3) in a way that can be handled effectively. We call methods based on (3) signal-domain approaches.

In this paper, we propose the first exact maximum likelihood estimator for line spectral estimation in the signal domain. We formulate (3) as a novel rank-constrained structured matrix recovery problem and solve it using the alternating direction method of multipliers (ADMM). The main contributions of this paper are summarized as follows:

- We discuss several possible attempts to solve the MLE in the signal domain, viz. previous low-rank Hankel or Toeplitz characterizations of the spectral sparsity constraint $\mathbf{x} \in S_0$ in (3), and highlight their limitations in terms of model accuracy, nonuniqueness of solution and the Cramér-Rao bound (CRB) (see Section II).
- We propose a novel rank-constrained Hankel-Toeplitz optimization method in the signal domain for the MLE and show its advantages in terms of equivalence to the MLE and uniqueness of solution (see Section III).
- We present an ADMM algorithm for the MLE in the signal domain based on the Hankel-Toeplitz method and analyze its convergence property. Similar ADMM algorithms are derived for the methods discussed in Section II for the sake of comparison (see Section IV).
- We generalize the proposed Hankel-Toeplitz and ADMM methods for the MLE from the full data to the missing data case and from the 1-D to the higher-dimensional cases (see Section V).
- We present extensive numerical results to demonstrate the better performance of the proposed approach with respect to global optimality and statistical efficiency as compared to existing parameter-domain, signal-domain and subspace methods (see Section VI).

A. Relations to Prior Art

To circumvent the nonconvexity of the MLE, subspace-based methods such as the multiple signal classification (MUSIC) [20] and the estimation of parameters by rotational invariant techniques (ESPRIT) [21] estimate the frequencies from the signal subspace of the sample covariance matrix. In the presence of multiple snapshots (consider for example direction-of-arrival estimation with a uniform linear array [1]), they are shown to be large-snapshot realizations of the MLE [22]. They are applicable to the single snapshot case, considered in this paper, using the

spatial smoothing preprocessing techniques [23], [24], [25], but only in the full data case.

An approximate maximum likelihood estimator has been proposed in [26] in the signal domain by relaxing the spectral sparsity constraint $\mathbf{x} \in S_0$ using a low-rank Hankel method. The Hankel method has strictly suboptimal performance, hence an improved double-Hankel (D-Hankel) method was presented in [27], [28]. But differently from the Hankel-Toeplitz method proposed in this paper, the D-Hankel method cannot precisely characterize the set S_0 . Interestingly, before [26], [27], [28], S_0 was precisely characterized via a low-rank positive-semidefinite (PSD) Toeplitz method in [4]. But to the best of our knowledge, no nonconvex algorithms have been proposed directly based on this Toeplitz method partly because it does not result in a unique Toeplitz solution, which causes convergence issues to any nonconvex algorithms (see Section II for details). In this paper, the nonuniqueness issue is completely solved in the proposed Hankel-Toeplitz method by means of an original combination of the previous Hankel and Toeplitz methods.

Using the constraint $\mathbf{x} \in S_0$ in (3) we actually control the spectral sparsity of signal $\mathbf{x}$, an idea that lies at the core of the so-called sparse methods that have dominated the research on line spectral estimation of the last two decades; see, e.g., the overview article [19]. In sparse methods, one usually swaps the objective function and the constraint in (3) and solves for the spectrally sparse signal $\mathbf{x}$ by optimizing a sparse metric in a feasible domain. To overcome the nonconvexity of the problem, convex relaxation approaches (known also as gridless sparse methods) have been developed based on the aforementioned Hankel or Toeplitz models, e.g., atomic norm (or total variation norm) methods [2], [4], [29], [30], [31], enhanced matrix completion [28], [32] and covariance fitting [30]. ADMM is also used to solve the resulting convex problems [29], [30]. These methods solve the line spectral estimation problem in the signal domain but do not solve the nonconvex MLE problem.

In the absence of noise, the proposed low-rank Hankel-Toeplitz method for the MLE is related to a rank minimization method in [33], where noiseless line spectral estimation was studied. However, the rank minimization problem in [33] is based on heuristics and its relation to the spectral sparsity is unclear. In contrast to this, we show in this paper that the minimal rank equals the spectral sparsity K if K is small, and the Hankel-Toeplitz method results in an equivalent reformulation of the (noisy) MLE and admits a unique solution. The key to our proof is the classical Carathéodory-Fejér theorem [1], Ch. 4.9.2 and a lemma regarding factorization of Hankel matrices [34]. Using a similar proof, we also extend the proposed Hankel-Toeplitz method from the 1-D to higher-dimensional line spectral estimation.

B. Notation

Notations used in this paper are as follows. Boldface letters are reserved for vectors and matrices. The set of real and complex numbers are denoted $\mathbb{R}$ and $\mathbb{C}$ respectively. For vector $\mathbf{x}$, $\mathbf{x}^T$, $\bar{\mathbf{x}}$, $\mathbf{x}^H$, $\|\mathbf{x}\|_1$ and $\|\mathbf{x}\|_2$ denote its transpose, complex conjugate, conjugate transpose, ℓ_1 norm and ℓ_2 norm, respectively. The

notation $|\cdot|$ denotes the amplitude of a scalar or cardinality of a set. Let $\Re\{\cdot\}$ and $\Im\{\cdot\}$ denote the real part and the imaginary part of a complex number, respectively. The matrix inverse, Moore-Penrose pseudo-inverse, complex conjugate, transpose and conjugate transpose of matrix $\mathbf{X}$ are denoted $\mathbf{X}^{-1}$, $\mathbf{X}^\dagger$, $\overline{\mathbf{X}}$, $\mathbf{X}^T$ and $\mathbf{X}^H$ respectively. The rank of matrix $\mathbf{X}$ is denoted $\text{rank}(\mathbf{X})$. $\text{range}(\mathbf{X})$ and $\text{tr}(\mathbf{X})$ mean the column space and the trace of $\mathbf{X}$, respectively. The inner product of matrices $\mathbf{X}$ and $\mathbf{Y}$ is defined as $\langle \mathbf{X}, \mathbf{Y} \rangle_{\mathbb{R}} = \Re\{\text{tr}(\mathbf{X}^H \mathbf{Y})\}$. $\mathbf{X} \geq \mathbf{0}$ means that $\mathbf{X}$ is Hermitian PSD. The diagonal matrix with vector $\mathbf{x}$ on the diagonal is denoted $\text{diag}(\mathbf{x})$. The j th entry of vector $\mathbf{x}$ is x_j and the (i, j) th entry of $\mathbf{X}$ is X_{ij} . An identity matrix is denoted as $\mathbf{I}$.

II. PREVIOUS LOW-RANK METHODS AND TWO CRBS

In this section, we present several previous attempts to solve the MLE (3) in the signal domain based on low-rank Hankel or Toeplitz characterizations of the spectral sparsity constraint $\mathbf{x} \in S_0$. Without loss of generality, we assume that the length N of regularly spaced samples $\{y_j\}$ is odd with $N = 2n - 1$ for integer n . If N is even, then we may assume that the $(N + 1)$ st sample is missing, which will be tackled in Section V.

A. Low-Rank Hankel Methods

A low-rank Hankel method is introduced in [26], [32] inspired by the classical Ho-Kalman algorithm and the matrix pencil approach [35], [36], [37]. In particular, for any $\mathbf{x} \in S_0$, we can form an $n \times n$ Hankel matrix $\mathcal{H}\mathbf{x}$ with its (i, j) entry given by x_{i+j-1} for $i, j \in \{1, \dots, n\}$. It can easily be shown that

$$\mathcal{H}\mathbf{x} = \mathbf{A}_n(\mathbf{f}) \mathbf{S} \mathbf{A}_n^T(\mathbf{f}) \quad (5)$$

where $\mathbf{A}_n(\mathbf{f}) = [\mathbf{a}_n(f_1), \dots, \mathbf{a}_n(f_K)]$ is an $n \times K$ Vandermonde matrix with $\mathbf{a}_n(f_k) = [1, e^{i2\pi f_k}, \dots, e^{i2\pi f_k(n-1)}]^T$ and $\mathbf{S} = \text{diag}(\mathbf{s})$ where $\mathbf{s}$ is the vector of $\{s_k\}$. Let

$$S_H = \{\mathbf{x} : \text{rank}(\mathcal{H}\mathbf{x}) \leq K\}, \quad (6)$$

$$S'_H = \left\{ \mathbf{x} : x_j = \sum_{k=1}^K s_k e^{(i2\pi f_k - \tau_k)(j-1)}, \right. \\ \left. f_k \in [0, 1), s_k \in \mathbb{C}, \tau_k \in \mathbb{R} \right\}. \quad (7)$$

It can easily be shown that

$$S_0 \subset S'_H \subset S_H \quad (8)$$

Moreover, by applying the Kronecker's theorem it is shown in [28] that if $K < n$, then $S_H \approx S'_H$ except for degenerate cases. The following rank-constrained *Hankel* optimization model is presented in [26] by relaxing S_0 to S_H :

$$(\text{Hankel}) \quad \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \text{subject to } \text{rank}(\mathcal{H}\mathbf{x}) \leq K. \quad (9)$$

To tighten the relaxation above, the set

$$S_{DH} = \{\mathbf{x} : \text{rank}[\mathcal{H}\mathbf{x} \mid \mathbf{J}\mathcal{H}\overline{\mathbf{x}}\mathbf{J}] \leq K\} \quad (10)$$

is proposed in [28], where $\mathbf{J}$ is an $n \times n$ reversal matrix, resulting in the following *D-Hankel* optimization model:

$$(\text{D-Hankel}) \quad \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{x}\|_2^2, \\ \text{subject to } \text{rank}([\mathcal{H}\mathbf{x} \mid \mathbf{J}\mathcal{H}\overline{\mathbf{x}}\mathbf{J}]) \leq K. \quad (11)$$

Define the set

$$S'_{DH} = \{\mathbf{x} \in S'_H \text{ where either } \tau_k = 0 \text{ or } \pm \tau_k \text{ appear in pair}\}. \quad (12)$$

It is shown in [28] that

$$S_0 \subset S'_{DH} \approx S_{DH} \subset S_H \quad (13)$$

Evidently, due to the relaxation from S_0 to S_H or S_{DH} , the aforementioned Hankel and D-Hankel models in (9) and (11) are not equivalent to the MLE in (3). In the presence of noise, the Hankel solution results in nonzero $\{\tau_k\}$ with probability one, while the D-Hankel solution might contain symmetric pairs $\{\pm\tau_k\}$ [28].

B. Low-Rank Toeplitz Methods

Let

$$\mathcal{T}\mathbf{t} = \begin{bmatrix} t_1 & t_2 & \dots & t_N \\ \bar{t}_2 & t_1 & \dots & t_{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{t}_N & \bar{t}_{N-1} & \dots & t_1 \end{bmatrix} \quad (14)$$

denote an $N \times N$ Hermitian Toeplitz matrix formed by using $\mathbf{t} \in \mathbb{C}^N$. It is shown in [4] that S_0 is identical to the set

$$S_T = \left\{ \mathbf{x} : \begin{bmatrix} u & \mathbf{x}^H \\ \mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \geq \mathbf{0}, \text{rank} \begin{bmatrix} u & \mathbf{x}^H \\ \mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \leq K \right. \\ \left. \text{for some } u \in \mathbb{R}, \mathbf{t} \in \mathbb{C}^N \right\}. \quad (15)$$

To see this, note that if $\mathbf{x} \in S_0$, then we may choose

$$\mathcal{T}\mathbf{t} = \sum_{k=1}^K p_k \mathbf{a}(f_k) \mathbf{a}^H(f_k), \quad u = \sum_{k=1}^K \frac{|s_k|^2}{p_k}, \quad p_k > 0 \quad (16)$$

such that $\mathbf{x} \in S_T$. Conversely, if $(\mathbf{x}, u, \mathcal{T}\mathbf{t})$ satisfies (15), then it is evident that $\mathcal{T}\mathbf{t} \geq \mathbf{0}$ and $\text{rank}(\mathcal{T}\mathbf{t}) \leq K$, and by applying the Carathéodory-Fejér theorem (or the Vandermonde decomposition lemma) [1], Ch. 4.9.2] we have

$$\mathcal{T}\mathbf{t} = \sum_{k=1}^K p_k \mathbf{a}(f_k) \mathbf{a}^H(f_k), \quad p_k \geq 0. \quad (17)$$

It then follows that $\mathbf{x} \in S_0$ since $\mathbf{x}$ lies in the range space of $\mathcal{T}\mathbf{t}$ by the first constraint in (15) and the column inclusion property of PSD matrices.

Substituting $S_0 = S_T$ into (3), the MLE problem is written equivalently as the following rank-constrained Toeplitz optimization model (referred to as *Toeplitz-I*):

$$(\text{Toeplitz-1}) \min_{\mathbf{x}, u, \mathbf{t}} \|\mathbf{y} - \mathbf{x}\|_2^2,$$

$$\text{subject to } \begin{bmatrix} u & \mathbf{x}^H \\ \mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \geq \mathbf{0}, \text{ rank } \begin{bmatrix} u & \mathbf{x}^H \\ \mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \leq K. \quad (18)$$

Seemingly the problem in (18) is a perfect rank-constrained structured matrix recovery formulation of the MLE; however, to the best of our knowledge, no algorithms based on it have been proposed. We will explain why this is so by describing below some difficulties encountered with (18).

Note first that a scaling ambiguity exists between u and $\mathcal{T}\mathbf{t}$. In particular, if $(\mathbf{x}, u, \mathbf{t})$ is a solution to (18), then so is $(\mathbf{x}, cu, c^{-1}\mathbf{t})$ for any $c > 0$. This issue is partly solved in [38] by imposing an extra constraint $u = t_1$, which leads to the following equivalent optimization model (referred to as *Toeplitz-2*):

$$(\text{Toeplitz-2}) \min_{\mathbf{x}, \mathbf{t}} \|\mathbf{y} - \mathbf{x}\|_2^2,$$

$$\text{subject to } \begin{bmatrix} t_1 & \mathbf{x}^H \\ \mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \geq \mathbf{0}, \text{ rank } \begin{bmatrix} t_1 & \mathbf{x}^H \\ \mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \leq K. \quad (19)$$

However, the solution to (19) remains non-unique for $K > 1$. To see this, suppose that $\mathbf{x} \in S_0$ is fixed to an optimal solution to (19). Evidently, any $(u, \mathcal{T}\mathbf{t})$ satisfying (16) results in an optimal solution to (18), with $\{p_k > 0\}$ being free variables. It follows from (16) that $t_1 = \sum_{k=1}^K p_k$ and thus letting $u = t_1$ is equivalent to imposing that

$$\sum_{k=1}^K \frac{|s_k|^2}{p_k} = \sum_{k=1}^K p_k, \quad (20)$$

which reduces only one degree of freedom in choosing $\{p_k\}$. Consequently, if $K > 1$, then there are still infinitely many possibilities to choose $\{p_k\}$, resulting in infinitely many optimal solutions to (19). Even worse, we can let one of $\{p_k\}$ approach zero and some others approach infinity such that (20) still holds. Then the solution for $\mathbf{t}$ becomes unbounded. These non-uniqueness and unboundedness issues may lead to convergence issues of the algorithm used to solve (18) or (19), which will be shown via numerical simulations in Section VI.

C. Two CRBs

The CRB is a lower bound on the (co)variance of any unbiased estimator and plays an important role in line spectral estimation [39], [40]. In principle, the MLE, if globally solved, achieves the CRB asymptotically. Recall the data model in (1) and denote the set of (real-valued) parameters $\boldsymbol{\theta} = \{f_k, \Re\{s_k\}, \Im\{s_k\}\}_{k=1}^K$. Let $\text{FIM}_{\boldsymbol{\theta}}$ be the corresponding Fisher information matrix (FIM) that is PSD. Then the CRB matrix is given by

$$\text{CRB}_{\boldsymbol{\theta}} = \text{FIM}_{\boldsymbol{\theta}}^{-1} \quad (21)$$

Since both the Toeplitz-1 and Toeplitz-2 optimization models are equivalent to the original MLE problem and estimate the same parameters $\boldsymbol{\theta}$, the corresponding estimators, if globally solved, achieve the above CRB asymptotically.

Now consider the Hankel model in (9). Substituting $S_H \approx S'_H$ into (9), we can write (9) as (except for degenerate cases):

$$\min_{\{f_k, s_k, \tau_k\}} \sum_{j=1}^N \left| y_j - \sum_{k=1}^K s_k e^{(i2\pi f_k - \tau_k)(j-1)} \right|^2 \quad (22)$$

This corresponds exactly to the MLE by assuming that the data model in (1) depends on new unknown parameters $\{\tau_k\}$, which are usually referred to as damping factors since this data model arises in applications typically with $\tau_k > 0$; see, e.g., [41], [42]. Consequently, in this damped data model, the set of parameters to estimate becomes $\{\boldsymbol{\theta}, \boldsymbol{\tau}\}$, resulting in the FIM

$$\text{FIM}_{\{\boldsymbol{\theta}, \boldsymbol{\tau}\}} = \begin{bmatrix} \text{FIM}_{\boldsymbol{\theta}} & \text{FIM}_{\boldsymbol{\theta}, \boldsymbol{\tau}} \\ \text{FIM}_{\boldsymbol{\theta}, \boldsymbol{\tau}}^T & \text{FIM}_{\boldsymbol{\tau}} \end{bmatrix} \quad (23)$$

where $\text{FIM}_{\boldsymbol{\theta}, \boldsymbol{\tau}}$ denotes the cross term. As a result, we obtain a new CRB corresponding to $\boldsymbol{\theta}$ in this *damped* data model, denoted by $\text{CRB}_{\boldsymbol{\theta}}^d$, as the (1,1) block of $\text{FIM}_{\{\boldsymbol{\theta}, \boldsymbol{\tau}\}}^{-1}$ that is given by:

$$\text{CRB}_{\boldsymbol{\theta}}^d = (\text{FIM}_{\boldsymbol{\theta}} - \text{FIM}_{\boldsymbol{\theta}, \boldsymbol{\tau}} \cdot \text{FIM}_{\boldsymbol{\tau}}^{-1} \cdot \text{FIM}_{\boldsymbol{\theta}, \boldsymbol{\tau}}^T)^{-1} \quad (24)$$

It follows immediately that

$$\text{CRB}_{\boldsymbol{\theta}}^d \geq \text{FIM}_{\boldsymbol{\theta}}^{-1} = \text{CRB}_{\boldsymbol{\theta}} \quad (25)$$

Therefore, it is expected that the Hankel estimator, if globally solved, achieves asymptotically $\text{CRB}_{\boldsymbol{\theta}}^d$, rather than $\text{CRB}_{\boldsymbol{\theta}}$.

In the D-Hankel model in (11), the prior knowledge $\{\tau_k = 0\}$ is partly exploited. As a result, the D-Hankel estimator, if globally solved, could possibly do better than $\text{CRB}_{\boldsymbol{\theta}}^d$ and approach $\text{CRB}_{\boldsymbol{\theta}}$ asymptotically.

Our previous analysis can be extended to other algorithms. In particular, if an algorithm, e.g., ESPRIT, is applied to the damped data model without exploiting the knowledge $\{\tau_k = 0\}$, then it cannot do better than $\text{CRB}_{\boldsymbol{\theta}}^d$ asymptotically. In contrast to this, Unitary-ESPRIT [43], ESPRIT with forward-backward spatially smoothing (FBSS) and MUSIC can possibly be better than $\text{CRB}_{\boldsymbol{\theta}}^d$ by partly using the prior $\{\tau_k = 0\}$. These findings will be verified numerically in Section VI.

III. HANKEL-TOEPLITZ OPTIMIZATION MODEL

In this section, we consider the typical scenario $K < n$ and show that the set S_0 can be written equivalently as

$$S_{\text{HT}} = \left\{ \mathbf{x} : \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \geq \mathbf{0}, \text{ rank } \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \leq K \right. \\ \left. \text{for some } \mathbf{t} \in \mathbb{C}^n \right\} \quad (26)$$

where $\mathcal{H}\mathbf{x}$ and $\mathcal{T}\mathbf{t}$ are $n \times n$ Hankel and Hermitian Toeplitz matrices. In particular, we have the following theorem.

Theorem 1: Assume $K < n$. The following statements are true:

- 1) $S_{\text{HT}} = S_0$;
- 2) For any $\mathbf{x} = \sum_{k=1}^K \mathbf{a}(f_k) s_k \in S_{\text{HT}}$ with nonzero s_k 's, there exists a unique $\mathbf{t}$ in (26) given by

$$\mathcal{T}\mathbf{t} = \mathbf{A}_n(\mathbf{f}) \text{diag}(|s|) \mathbf{A}_n^H(\mathbf{f}), \quad (27)$$

where $|\cdot|$ operates on s element-wise.

Proof: We first show $S_0 \subset S_{HT}$. For any $\mathbf{x} = \sum_{k=1}^K \mathbf{a}(f_k) s_k \in S_0$, we have (5). Let $\mathcal{T}\mathbf{t}$ be as in (27) and denote $\mathbf{P} = \text{diag}(|s|)$. For notational simplicity, we write $\mathbf{A}_n(\mathbf{f})$ as $\mathbf{A}$ hereafter. It follows that

$$\begin{aligned} \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} &= \begin{bmatrix} \bar{\mathbf{A}}\mathbf{P}\mathbf{A}^T & \bar{\mathbf{A}}\bar{\mathbf{S}}\mathbf{A}^H \\ \mathbf{A}\mathbf{S}\mathbf{A}^T & \mathbf{A}\mathbf{P}\mathbf{A}^H \end{bmatrix} \\ &= \begin{bmatrix} \bar{\mathbf{A}} & \mathbf{0} \\ \mathbf{0} & \mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{P} & \bar{\mathbf{S}} \\ \mathbf{S} & \mathbf{P} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{A}} & \mathbf{0} \\ \mathbf{0} & \mathbf{A} \end{bmatrix}^H \\ &\geq \mathbf{0}. \end{aligned} \quad (28)$$

Moreover,

$$\text{rank} \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} = \text{rank} \begin{bmatrix} \mathbf{P} & \bar{\mathbf{S}} \\ \mathbf{S} & \mathbf{P} \end{bmatrix} = \text{rank}(\mathbf{P}) \leq K, \quad (29)$$

where the second equality holds since the Schur complement

$$\mathbf{P} - \bar{\mathbf{S}}\mathbf{P}^{-1}\mathbf{S} = \mathbf{0} \quad (30)$$

Consequently, we have $\mathbf{x} \in S_{HT}$, and thus $S_0 \subset S_{HT}$.

We next show $S_{HT} \subset S_0$. For any $\mathbf{x} \in S_{HT}$, it follows from (26) and the column inclusion property of PSD matrices that

$$\text{rank}(\mathcal{T}\mathbf{t}) \leq K, \quad \mathcal{H}\mathbf{x} \in \text{range}(\mathcal{T}\mathbf{t}). \quad (31)$$

By making use of the Vandermonde decomposition lemma [1], Ch. 4.9.2], there exist distinct $\{f_k\}_{k=1}^K$ and $\{p_k \geq 0\}_{k=1}^K$ such that

$$\mathcal{T}\mathbf{t} = \mathbf{A}\mathbf{P}\mathbf{A}^H, \quad \mathbf{P} = \text{diag}(\mathbf{p}). \quad (32)$$

It then follows from (31) that there exists $K \times n$ matrix $\mathbf{G}$ such that

$$\mathcal{H}\mathbf{x} = \mathbf{A}\mathbf{G} = \mathbf{G}^T \mathbf{A}^T, \quad (33)$$

where the second equality follows from the symmetry of $\mathcal{H}\mathbf{x}$. We further have $\mathbf{G} = \mathbf{A}^\dagger \mathbf{G}^T \mathbf{A}^T$ and thus

$$\mathcal{H}\mathbf{x} = \mathbf{A}\mathbf{S}'\mathbf{A}^T, \quad (34)$$

where $\mathbf{S}' = \mathbf{A}^\dagger \mathbf{G}^T$ is a $K \times K$ matrix. Given the above factorization of the Hankel matrix $\mathcal{H}\mathbf{x}$ with $K < n$, it follows from [34], Lemma 4] that $\mathbf{S}'$ must be diagonal, i.e. $\mathbf{S}' = \text{diag}(\mathbf{s})$, so that (34) is a Vandermonde decomposition. It follows immediately from (34) that

$$\mathbf{x} = \sum_{k=1}^K \mathbf{a}(f_k) s_k \in S_0, \quad (35)$$

completing the proof of the first part.

To show the second part, note first that $\text{rank}(\mathcal{H}\mathbf{x}) = \text{rank}(\mathbf{S}) = K$ since s_k 's are nonzero, which yields that $\text{rank}(\mathcal{T}\mathbf{t}) = K$ by applying (31). Consequently, we must have

$$(32) \text{ with } p_k > 0. \text{ Now consider the block matrix } \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix}$$

that still satisfies (28). Applying consecutively $p_k > 0$, (26) and (28), we have that

$$\text{rank}(\mathbf{P}) = K \geq \text{rank} \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} = \text{rank} \begin{bmatrix} \mathbf{P} & \bar{\mathbf{S}} \\ \mathbf{S} & \mathbf{P} \end{bmatrix} \quad (36)$$

This is true only if the Schur complement

$$\mathbf{P} - \bar{\mathbf{S}}\mathbf{P}^{-1}\mathbf{S} = \mathbf{0}, \quad (37)$$

or equivalently,

$$p_k = |s_k|, \quad k = 1, \dots, K. \quad (38)$$

Inserting (38) into (32) yields (27), completing the proof.

Making use of Theorem 1, the MLE problem in (3) is written equivalently as the following rank-constrained *Hankel-Toeplitz* optimization model:

$$\begin{aligned} (\text{Hankel-Toeplitz}) \quad & \min_{\mathbf{x}, \mathbf{t}} \|\mathbf{y} - \mathbf{x}\|_2^2, \\ & \text{subject to } \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \geq \mathbf{0}, \\ & \text{rank} \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \leq K. \end{aligned} \quad (39)$$

Suppose that the maximum likelihood estimator is given by $\{f_k^{\text{ML}}, s_k^{\text{ML}}\}_{k=1}^K$, where $s_k^{\text{ML}} \neq 0$ almost surely (otherwise, $\mathbf{y}$ would be composed exactly of $K - 1$ or fewer sinusoids, which occurs with probability zero in the presence of noise). Then, applying Theorem 1, an optimal solution to (39) is given by

$$\begin{aligned} \mathbf{x}^{\text{ML}} &= \sum_{k=1}^K \mathbf{a}(f_k^{\text{ML}}) s_k^{\text{ML}}, \\ \mathcal{T}\mathbf{t}^{\text{ML}} &= \mathbf{A}_n(f^{\text{ML}}) \text{diag}(|s^{\text{ML}}|) \mathbf{A}_n^H(f^{\text{ML}}), \end{aligned} \quad (40)$$

where $\mathbf{t}^{\text{ML}}$ is uniquely determined by $\mathbf{x}^{\text{ML}}$.

IV. ADMM ALGORITHMS

A. A General Algorithmic Framework

We have presented five candidate optimization models for the MLE in the signal domain: Hankel, D-Hankel, Toeplitz-1, Toeplitz-2 and Hankel-Toeplitz given in (9), (11), (18), (19) and (39), respectively. The three Toeplitz-based models are equivalent to the MLE. Moreover, the proposed Hankel-Toeplitz admits a unique solution. All five models are rank-constrained, which is the only part making them nonconvex and challenging to solve. Interestingly, the ADMM algorithm has been shown to be a good choice for nonconvex optimization problems such as non-negative tensor decomposition [44], matrix separation [45], and phase retrieval [46], especially for rank-constrained problems [26], [47], [48]. Theoretical progress has also been made in [49], [50], [51], [52], [53], [54]. In this paper, we use the ADMM to solve the five rank-constrained matrix recovery problems. Note that all five models can be written as:

$$\min_{\mathbf{z}} h(\mathbf{z}), \quad \text{subject to } \mathcal{M}\mathbf{z} \in \mathcal{S}_K, \quad (41)$$

where $\mathbf{z}$ denotes the set of variables to optimize and $\mathcal{M}\mathbf{z}$ denotes a matrix that is linear in (the real and complex parts of) $\mathbf{z}$. The

set $\mathcal{S}_K$ is composed of matrices with rank no greater than K for Hankel and D-Hankel, while the matrices are also Hermitian PSD for the other three models.

To apply the ADMM, we introduce an auxiliary matrix variable $\mathbf{Q}$ and rewrite (41) as

$$\min_{\mathbf{z}, \mathbf{Q}} h(\mathbf{z}) + \delta_{\mathcal{S}_K}(\mathbf{Q}), \text{ subject to } \mathbf{Q} = \mathcal{M}\mathbf{z}, \quad (42)$$

where $\delta_{\mathcal{S}_K}(\cdot)$ is the indicator function: $\delta_{\mathcal{S}_K}(\mathbf{Q}) = 0$ if $\mathbf{Q} \in \mathcal{S}_K$ or ∞ if $\mathbf{Q} \notin \mathcal{S}_K$. The augmented Lagrangian function is given by

$$\begin{aligned} \mathcal{L}(\mathbf{Q}, \mathbf{z}, \mathbf{\Lambda}) &= h(\mathbf{z}) + \delta_{\mathcal{S}_K}(\mathbf{Q}) + \langle \mathbf{\Lambda}, \mathbf{Q} - \mathcal{M}\mathbf{z} \rangle_{\mathbb{R}} + \frac{\mu}{2} \|\mathbf{Q} - \mathcal{M}\mathbf{z}\|_{\text{F}}^2 \\ &= h(\mathbf{z}) + \delta_{\mathcal{S}_K}(\mathbf{Q}) + \frac{\mu}{2} \|\mathbf{Q} - \mathcal{M}\mathbf{z} + \mu^{-1}\mathbf{\Lambda}\|_{\text{F}}^2 - \frac{1}{2\mu} \|\mathbf{\Lambda}\|_{\text{F}}^2, \end{aligned} \quad (43)$$

where μ is the penalty parameter and $\mathbf{\Lambda}$ is the Hermitian Lagrangian multiplier. The ADMM algorithm iterates the following three steps:

$$\mathbf{Q}^{l+1} = \arg \min_{\mathbf{Q} \in \mathcal{S}_K} \|\mathbf{Q} - \mathcal{M}\mathbf{z}^l + \mu^{-1}\mathbf{\Lambda}^l\|_{\text{F}}^2, \quad (44)$$

$$\mathbf{z}^{l+1} = \arg \min_{\mathbf{z}} h(\mathbf{z}) + \frac{\mu}{2} \|\mathbf{Q}^{l+1} - \mathcal{M}\mathbf{z} + \mu^{-1}\mathbf{\Lambda}^l\|_{\text{F}}^2, \quad (45)$$

$$\mathbf{\Lambda}^{l+1} = \mathbf{\Lambda}^l + \mu (\mathbf{Q}^{l+1} - \mathcal{M}\mathbf{z}^{l+1}), \quad (46)$$

where l denotes the iteration index. It follows from (44) that

$$\mathbf{Q}^{l+1} = \mathcal{P}_{\mathcal{S}_K}(\mathcal{M}\mathbf{z}^l - \mu^{-1}\mathbf{\Lambda}^l), \quad (47)$$

the projection of $\mathcal{M}\mathbf{z}^l - \mu^{-1}\mathbf{\Lambda}^l$ onto the set $\mathcal{S}_K$. In (45), $\mathbf{z}$ can usually be solved for in closed form.

The ADMM iterates are terminated if the primal and dual residuals computed with

$$\mathbf{R}_p^{l+1} = \mathbf{Q}^{l+1} - \mathcal{M}\mathbf{z}^{l+1}, \quad \mathbf{R}_d^{l+1} = \mu (\mathcal{M}\mathbf{z}^{l+1} - \mathcal{M}\mathbf{z}^l)$$

are sufficiently small or a maximum number of iterations is reached; see [47]. A varying penalty parameter μ is also used as in [47], 3.4.1 to accelerate convergence.

Once the ADMM algorithm has converged, the Toeplitz or Hankel matrix involved is expected to have rank K and admit the Vandermonde decomposition in (17) or (5). Therefore, we compute the corresponding Vandermonde decomposition by applying the ESPRIT algorithm, from which the estimated frequencies $\{\hat{f}_k\}$ are obtained. After that, the estimated coefficients $\{\hat{s}_k\}$ are computed as:

$$\hat{\mathbf{s}} = \mathbf{A}^\dagger(\hat{\mathbf{f}})\mathbf{y}. \quad (48)$$

We note that $\hat{\mathbf{s}}$ can also be obtained from the equality $\hat{\mathbf{x}} = \mathbf{A}(\hat{\mathbf{f}})\hat{\mathbf{s}}$.

B. Details of ADMM Updates

We derive the detailed ADMM updates for the five signal-domain methods in this subsection and Appendix VII-A. The key is to identify $\mathbf{z}$, $\mathcal{M}\mathbf{z}$ and to compute $\mathbf{z}$, $\mathbf{Q}$ in (47) and (45).

We first consider the proposed Hankel-Toeplitz model in (39) and make the substitutions

$$\mathbf{z} = \{\mathbf{x}, \mathbf{t}\}, \quad h(\mathbf{z}) = \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \mathcal{M}\mathbf{z} = \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix}. \quad (49)$$

Write $\mathbf{Q} = \begin{bmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{21}^H \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{bmatrix}$ and $\mathbf{\Lambda} = \begin{bmatrix} \mathbf{\Lambda}_{11} & \mathbf{\Lambda}_{21}^H \\ \mathbf{\Lambda}_{21} & \mathbf{\Lambda}_{22} \end{bmatrix}$ as block matrices like $\mathcal{M}\mathbf{z}$. It follows from (44) that $\mathbf{Q}$ is updated as:

$$\mathbf{Q}^{l+1} = \mathcal{P}_{\mathcal{S}_K} \left(\begin{bmatrix} \mathcal{T}\bar{\mathbf{t}}^l & \mathcal{H}\bar{\mathbf{x}}^l \\ \mathcal{H}\mathbf{x}^l & \mathcal{T}\mathbf{t}^l \end{bmatrix} - \mu^{-1} \begin{bmatrix} \mathbf{\Lambda}_{11}^l & (\mathbf{\Lambda}_{21}^l)^H \\ \mathbf{\Lambda}_{21}^l & \mathbf{\Lambda}_{22}^l \end{bmatrix} \right), \quad (50)$$

where $\mathcal{S}_K$ is the set of Hermitian PSD matrices of rank no greater than K . The projection is obtained as the truncated eigen-decomposition of the Hermitian matrix argument by setting all but the largest K positive eigenvalues to zero (note that if the number of positive eigenvalues is smaller than K , then just keep them and set the others to zero).

Since the variables $\mathbf{x}$ and $\mathbf{t}$ are separable, equating the derivative of the objective function in (45) with respect to $\{\mathbf{x}, \mathbf{t}\}$ to zero yields the updates

$$\mathbf{x}^{l+1} = (\mathbf{I} + \mu \mathbf{D}_{\mathcal{H}})^{-1} [\mathbf{y} + \mu \mathcal{H}^H (\mathbf{Q}_{21}^{l+1} + \mu^{-1}\mathbf{\Lambda}_{21}^l)], \quad (51)$$

$$\mathbf{t}^{l+1} = \frac{1}{2} \mathbf{D}_{\mathcal{T}}^{-1} \mathcal{T}^H \left(\overline{\mathbf{Q}_{11}^{l+1}} + \mathbf{Q}_{22}^{l+1} + \mu^{-1} (\overline{\mathbf{\Lambda}_{11}^l} + \mathbf{\Lambda}_{22}^l) \right) \quad (52)$$

where $\mathcal{H}^H$ denotes the adjoint of the Hankel operator $\mathcal{H}$ by mapping an $n \times n$ matrix $\mathbf{X}$ to an N -dimensional vector with $\mathcal{H}^H \mathbf{X} = \left\{ \sum_{i+j-1=l} X_{ij} \right\}_{l=1}^N$, $\mathbf{D}_{\mathcal{H}} = \text{diag}\{1, 2, \dots, n, n-1, \dots, 1\}$, $\mathcal{T}^H$ denotes the adjoint of the Toeplitz operator $\mathcal{T}$ by mapping an $n \times n$ Hermitian matrix $\mathbf{X}$ to an n -dimensional vector $\mathcal{T}^H \mathbf{X} = \left\{ \sum_{j-i=l-1} X_{ij} \right\}_{l=1}^n$, and $\mathbf{D}_{\mathcal{T}} = \text{diag}\{n, n-1, \dots, 1\}$. The unique stationary point $\{\mathbf{x}, \mathbf{t}\}$ given above is the minimizer to (45).

According to (46), $\mathbf{\Lambda}$ is updated as:

$$\mathbf{\Lambda}^{l+1} = \mathbf{\Lambda}^l + \mu \left(\mathbf{Q}^{l+1} - \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}}^{l+1} & \mathcal{H}\bar{\mathbf{x}}^{l+1} \\ \mathcal{H}\mathbf{x}^{l+1} & \mathcal{T}\mathbf{t}^{l+1} \end{bmatrix} \right). \quad (53)$$

The details of the updates for the other four problems, which can be similarly derived, are presented in Appendix VII-A.

Note that the computations of all ADMM algorithms are dominated by the projection $\mathcal{P}_{\mathcal{S}_K}$ in each iteration that is computed either by the truncated eigen-decomposition or the truncated singular value decomposition (SVD), which takes $O(N^3)$ flops by computing the full decompositions and can be accelerated using advanced techniques [55].

C. Convergence

The global convergence of the ADMM for convex optimization problems is well studied and understood [47]. For certain nonconvex problems, its (subsequential) convergence to a stationary point has been studied in [49], [50], [51], [52], [53], [54]. Inspired by the results in the cited papers, we state the

following result for the proposed ADMM algorithm based on the Hankel-Toeplitz model.

Theorem 2: Let $\{Q^l, z^l, \Lambda^l\}$ be a sequence generated by the ADMM for the Hankel-Toeplitz model. Assume that

$$\lim_{l \rightarrow \infty} \|z^{l+1} - z^l\|_2^2 + \|\Lambda^{l+1} - \Lambda^l\|_F^2 = 0. \quad (54)$$

Then for any limit point $\{Q^*, z^*, \Lambda^*\}$, z^* is a stationary point of (42), i.e.,

$$\begin{aligned} 0 &\in \partial \delta_{S_K}(Q^*) + \Lambda^*, \\ \nabla h(z^*) &= \mathcal{M}^H \Lambda^*, \\ Q^* &= \mathcal{M} z^*, \end{aligned} \quad (55)$$

where $\partial \delta_{S_K}$ is the general subgradient [56], Definition 8.3].

Proof: See Appendix VII-B.

Theorem 2 states that if the sequence $\{Q^l, z^l, \Lambda^l\}$ does not exhibit any jumps (see (54)), then any limit point is a stationary point of the primal problem (42). In other words, whenever the sequence converges, it converges to a stationary point. Assumptions similar to (54) have been made in [49], [50]. It is shown in [51], [52], [53], [54] that such assumptions can be weakened if, in the language of this paper, $\mathcal{M}$ is surjective; however, this is not true for our problem.

V. EXTENSIONS

In this section, we extend the proposed Hankel-Toeplitz optimization model and the corresponding ADMM algorithm to the missing data case and to 2-D and higher-dimensional line spectral estimation.

A. The Missing Data Case

In the missing data case, only samples supported on the index set $\Omega \subseteq \{1, \dots, N\}$, of size $M < N$, are observed. The observed samples are given by the $N \times 1$ vector $\mathbf{y}_\Omega = \mathcal{P}_\Omega(\mathbf{y})$, where $\mathcal{P}_\Omega$ is the projection onto the subspace supported on Ω that sets all entries outside of Ω to zero. Under the assumption of i.i.d. circular complex Gaussian noise, the MLE is formulated in the signal domain as:

$$\min_{\mathbf{x} \in \mathbb{C}^N} \|\mathcal{P}_\Omega(\mathbf{y} - \mathbf{x})\|_2^2, \text{ subject to } \mathbf{x} \in S_0. \quad (56)$$

Using the Hankel-Toeplitz method, the MLE is written equivalently as

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{t}} \|\mathcal{P}_\Omega(\mathbf{y} - \mathbf{x})\|_2^2, \\ \text{subject to } \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \geq \mathbf{0}, \text{ rank } \begin{bmatrix} \mathcal{T}\bar{\mathbf{t}} & \mathcal{H}\bar{\mathbf{x}} \\ \mathcal{H}\mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix} \leq K. \end{aligned} \quad (57)$$

The ADMM algorithm can be derived similarly as in the full data case. The only difference is the update of $\mathbf{x}$:

$$\begin{aligned} \mathbf{x}^{l+1} \\ = (\text{diag}(\mathcal{P}_\Omega(\mathbf{1})) + \mu \mathbf{D}_H)^{-1} [\mathbf{y}_\Omega + \mu \mathcal{H}^H (Q_{21}^l + \mu^{-1} \Lambda_{21}^l)] \end{aligned}$$

where $\mathbf{1}$ is the $N \times 1$ vector of ones. Note that similar modifications can also be made for the other ADMM algorithms to make them usable in this case.

B. Higher-Dimensional Line Spectral Estimation

The problem of higher-dimensional (such as 2-D, etc) line spectral estimation arises in radar and wireless communications [1], [57], [58]. We focus on the 2-D case for clarity. Certain notations such as $\mathbf{A}$, S_0 and S_{HT} have to be modified. The 2-D samples $\{Y_{j_1 j_2}\}$ are given by

$$\begin{aligned} Y_{j_1 j_2} &= X_{j_1 j_2} + \varepsilon_{j_1 j_2} \\ &= \sum_{k=1}^K s_k e^{i2\pi f_{k,1}(j_1-1)} e^{i2\pi f_{k,2}(j_2-1)} + \varepsilon_{j_1 j_2} \end{aligned} \quad (58)$$

for $j_1 = 1, \dots, N_1$, $j_2 = 1, \dots, N_2$ where $\mathbf{f}_k \triangleq [f_{k,1}, f_{k,2}] \in [0, 1)^2$, $k = 1, \dots, K$ are distinct frequencies.

As in the 1-D case, we assume that N_1, N_2 are odd and let $n_d = (N_d + 1)/2$ for $d = 1, 2$. Let $\mathbf{X}_{j_1,:}$ denote the j_1 -th row of $\mathbf{X}$, based on which an $n_2 \times n_2$ Hankel matrix $\mathcal{H}\mathbf{X}_{j_1,:}$ is formed. All these Hankel matrices are used to form a 2-level Hankel matrix $\mathcal{H}\mathbf{X} \in \mathbb{C}^{n_1 n_2 \times n_1 n_2}$ given by

$$\mathcal{H}\mathbf{X} = \begin{bmatrix} \mathcal{H}\mathbf{X}_{1,:} & \mathcal{H}\mathbf{X}_{2,:} & \dots & \mathcal{H}\mathbf{X}_{n_1,:} \\ \mathcal{H}\mathbf{X}_{2,:} & \mathcal{H}\mathbf{X}_{3,:} & \dots & \mathcal{H}\mathbf{X}_{n_1+1,:} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{H}\mathbf{X}_{n_1,:} & \mathcal{H}\mathbf{X}_{n_1+1,:} & \dots & \mathcal{H}\mathbf{X}_{N_1,:} \end{bmatrix}.$$

Moreover, for a 2-D complex array $\mathbf{T} \in \mathbb{C}^{n_1 \times N_2}$, we form an $n_2 \times n_2$ Toeplitz matrix $\mathcal{T}\mathbf{T}_{j_1,:}$ (non-Hermitian except for $\mathcal{T}\mathbf{T}_{1,:}$) by using $\mathbf{T}_{j_1,:}$. All such Toeplitz matrices are used to form a Hermitian 2-level Toeplitz matrix $\mathcal{T}\mathbf{T} \in \mathbb{C}^{n_1 n_2 \times n_1 n_2}$ given by

$$\mathcal{T}\mathbf{T} = \begin{bmatrix} \mathcal{T}\mathbf{T}_{1,:} & \mathcal{T}\mathbf{T}_{2,:} & \dots & \mathcal{T}\mathbf{T}_{n_1,:} \\ (\mathcal{T}\mathbf{T}_{2,:})^H & \mathcal{T}\mathbf{T}_{1,:} & \dots & \mathcal{T}\mathbf{T}_{n_1-1,:} \\ \vdots & \vdots & \ddots & \vdots \\ (\mathcal{T}\mathbf{T}_{n_1,:})^H & (\mathcal{T}\mathbf{T}_{n_1-1,:})^H & \dots & \mathcal{T}\mathbf{T}_{1,:} \end{bmatrix}.$$

For $d = 1, 2$, we define $\mathbf{A}_d$ as an $n_d \times K$ Vandermonde matrix with $[\mathbf{A}_d]_{j,k} = e^{i2\pi f_{k,d}(j-1)}$ and $\mathbf{a}_d(f_{k,d})$ as the k -th column of $\mathbf{A}_d$. Let $\mathbf{A} = \mathbf{A}_1 \star \mathbf{A}_2 \in \mathbb{C}^{n_1 n_2 \times K}$, where $\star$ denotes the Khatri-Rao (or column-wise Kronecker) product.

Using the above definitions, we redefine

$$\begin{aligned} S_0 &= \left\{ \sum_{k=1}^K s_k \mathbf{a}_1(f_{k,1}) \mathbf{a}_2^T(f_{k,2}) : \mathbf{f}_k \in [0, 1)^2, s_k \in \mathbb{C} \right\}, \\ S_{HT} &= \left\{ \mathbf{X} : \begin{bmatrix} \mathcal{T}\bar{\mathbf{T}} & \mathcal{H}\bar{\mathbf{X}} \\ \mathcal{H}\mathbf{X} & \mathcal{T}\mathbf{T} \end{bmatrix} \geq \mathbf{0}, \text{ rank } \begin{bmatrix} \mathcal{T}\bar{\mathbf{T}} & \mathcal{H}\bar{\mathbf{X}} \\ \mathcal{H}\mathbf{X} & \mathcal{T}\mathbf{T} \end{bmatrix} \leq K \right. \\ &\quad \left. \text{for some } \mathbf{T} \in \mathbb{C}^{n_1 \times N_2} \right\}. \end{aligned} \quad (59)$$

Then, we have the following corollary to Theorem 1.

Corollary 1: Assume $K < \min\{n_1, n_2\}$. The following statements are true:

- 1) $S_{HT} = S_0$;
- 2) For any $\mathbf{X} = \sum_{k=1}^K s_k \mathbf{a}_1(f_{k,1}) \mathbf{a}_2^T(f_{k,2})$ with nonzero s_k 's, there exists a unique $\mathbf{T}$ in (59) given by

$$\mathcal{T}\mathcal{T} = \mathbf{A} \text{diag}(|s|) \mathbf{A}^H. \quad (60)$$

Proof: We can follow the same steps as in the proof of Theorem 1. The key is to show that both the 2-level Toeplitz and Hankel matrices $\mathcal{T}\mathcal{T}$, $\mathcal{H}\mathbf{X}$ in (59) admit Vandermonde decompositions as in (32), (34). But differently from the previous proof, we cannot use the classical Carathéodory-Fejér theorem and [34], Lemma 4] since they apply only to the 1-D case. Fortunately, these results have been generalized to the 2-D case (in fact, to arbitrary dimensions) in [34], Theorem 1] and [34], Lemma 3] under the condition that $\text{rank}(\mathcal{T}\mathcal{T})$ or the number of columns of $\mathbf{A}$ in (34) is less than $\min\{n_1, n_2\}$, which is easy to verify from (59). Completing the proof using these extended results is more or less straightforward, and we omit the details. ■

Making use of Corollary 1, we can formulate the MLE for 2-D line spectral estimation as the following rank-constrained Hankel-Toeplitz optimization problem:

$$\begin{aligned} \min_{\mathbf{X}, \mathcal{T}} \|\mathbf{Y} - \mathbf{X}\|_F^2, \\ \text{subject to } \begin{bmatrix} \mathcal{T}\bar{\mathcal{T}} & \mathcal{H}\bar{\mathbf{X}} \\ \mathcal{H}\mathbf{X} & \mathcal{T}\mathcal{T} \end{bmatrix} \geq \mathbf{0}, \text{rank} \begin{bmatrix} \mathcal{T}\bar{\mathcal{T}} & \mathcal{H}\bar{\mathbf{X}} \\ \mathcal{H}\mathbf{X} & \mathcal{T}\mathcal{T} \end{bmatrix} \leq K, \end{aligned}$$

where the solution to $\mathbf{T}$ is uniquely determined by $\mathbf{X}$. Consequently, an ADMM algorithm can be derived for this problem as in the 1-D case. When the algorithm has converged, the 2-D frequencies can be retrieved from $\mathcal{T}\mathcal{T}$ by using the matrix pencil and pairing (MaPP) method in [34].

We note that our techniques and results can be extended to higher dimensions by composing the multi-level Hankel and Hermitian Toeplitz matrices recursively. The extended techniques are usable in the missing data case with minor modifications as in the previous subsection.

We remark that the assumption $K < \min\{n_1, n_2\}$ in Corollary 1 is not a limit for the proposed method. In fact, if it is not satisfied, we can always consider $\mathbf{Y}$ as incomplete observations of an enlarged $N'_1 \times N'_2$ virtual dataset $\mathbf{Y}'$ with $K < \min\{n'_1, n'_2\}$, where n'_1, n'_2 are defined similarly to n_1, n_2 , and then solve the problem as in the missing data case.

VI. NUMERICAL RESULTS

In this section, we carry out extensive numerical simulations to evaluate the performance of the proposed Hankel-Toeplitz-ADMM approach with respect to convergence, global optimality and statistical efficiency. We compare the proposed approach with the other ADMM methods based on Toeplitz-1, Toeplitz-2, Hankel and D-Hankel, which together with the proposed one are abbreviated as T1, T2, H, DH and HT respectively. In ADMM, the penalty parameter μ is initialized by $\mu_0 = 1/\sqrt{N}$ and varies as in [47], 3.4.1]. The ADMM will be terminated if a maximum number of 2000 iterations is reached. We also select the state-of-the-art parameter-domain method NOMP [18] for

comparison, which is a greedy algorithm and does not require an explicit initialization. We feed into NOMP the true model order K and set the over-sampling factor $\gamma = 4$, the Newton step $R_s = 1$ in the single refinement step, and the number of rounds $R_c = 50$ in cyclic refinements; see the cited paper for details.

A. Convergence and Global Optimality

We evaluate convergence and global optimality of the proposed approach in this subsection. The true global optimizer $\mathbf{f}^{\text{ML}}$ for the frequencies is found via a K -dimensional grid search (with a uniform grid of size 10^3 for each dimension) followed by a Newton's method. We consider that the global optimality is achieved by an approach if the distance of the estimated frequencies from $\mathbf{f}^{\text{ML}}$, measured by the root mean squared error (RMSE) $\|\hat{\mathbf{f}} - \mathbf{f}^{\text{ML}}\|_2/\sqrt{K}$, is less than 10^{-3} .

In *Experiment 1* we consider the full data case, with varying frequency separation and signal to noise ratio (SNR). In particular, we consider $K = 2$ spectral frequencies that are randomly generated such that they are separated by exactly Δ_f . The coefficients $\{s_k\}$ are generated with amplitudes $0.5 + |\alpha|$ and random phases, where α follows the standard normal distribution. A number of $N = 65$ uniform samples are generated containing white complex Gaussian noise, and the SNR is defined by $10 \log_{10}(\|\mathbf{x}^*\|_2^2/\|\boldsymbol{\varepsilon}\|_2^2)$ where $\mathbf{x}^*$ is the noiseless signal. We consider $\Delta_f \cdot N \in \{0.1, 0.4, 0.7, 1.0, 2.0\}$ and $\text{SNR} \in \{0, 10, \dots, 40\}$ dB. A number of 20 Monte Carlo runs are carried out for each combination of Δ_f and SNR.

The averaged number of iterations is shown in Fig. 1 with respect to (SNR, Δ_f) for the five ADMM algorithms. It is seen that T1 suffers from the convergence issue in the low SNR regime, while the number of iterations of T2 is sensitive to the frequency separation. In contrast to T1 and T2, the convergence is greatly accelerated for the proposed HT thanks to uniqueness of its solution. HT tends to require more iterations in presence of closely located frequencies, like T1 and T2. The H and DH have faster convergence partly because no Toeplitz matrices and thus less optimization variables are involved in them.

The success rates of achieving global optimality are presented in Fig. 2 where white means complete success and black means complete failure. It is seen that the proposed HT achieves global optimality with high probability, improves upon T1 and T2 and outperforms NOMP. The DH also has good performance in this full data case and improves upon H.

In *Experiment 2* we consider the missing data case, with varying sample size and SNR. A number of $K = 2$ frequencies are randomly generated, and the coefficients $\{s_k\}$ are randomly generated as before. A number of M samples are randomly selected from $N = 65$ regularly spaced noisy samples. We consider sample size $M \in \{20, 30, \dots, 60\}$ and, as before, $\text{SNR} \in \{0, 10, \dots, 40\}$ dB.

The averaged number of iterations is presented in Fig. 3. Again, it is seen that the convergence of HT is greatly accelerated as compared to T1 and T2. The number of iterations of HT is close to that of H and DH in this missing data case.

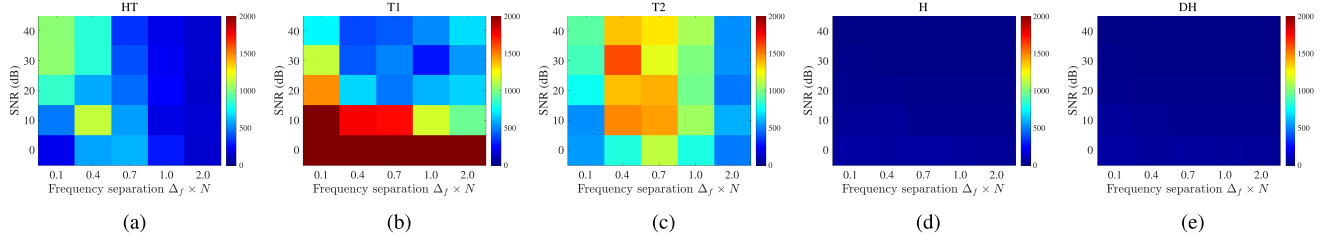


Fig. 1. Averaged numbers of iterations of (a) HT, (b) T1, (c) T2, (d) H, (e) DH in the full data case, with varying frequency separation and SNR.

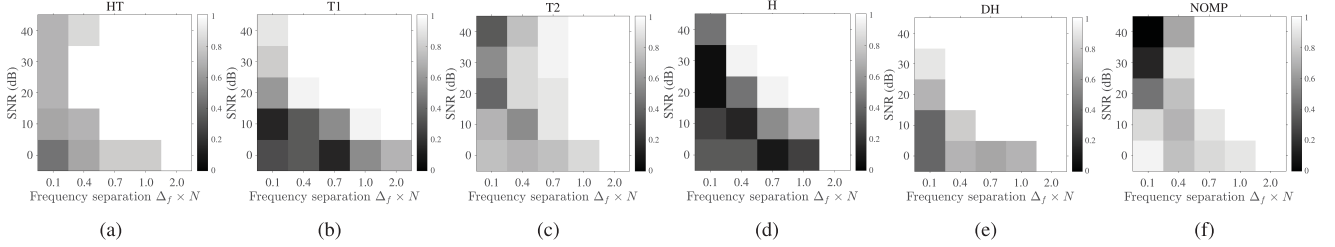


Fig. 2. Success rates of achieving global optimality of (a) HT, (b) T1, (c) T2, (d) H, (e) DH, (f) NOMP in the full data case, with varying frequency separation and SNR.

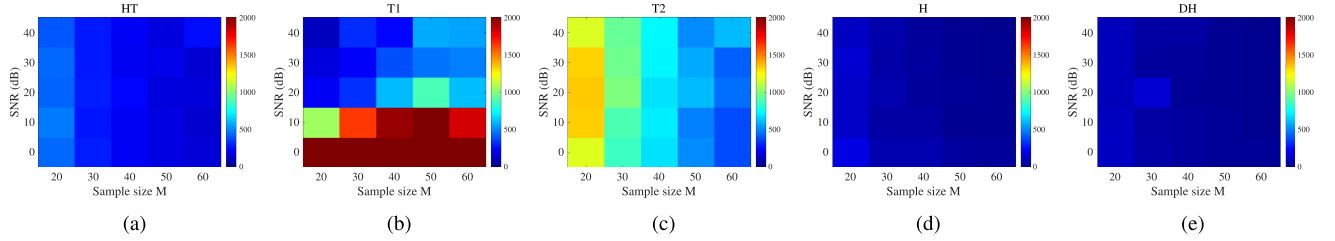


Fig. 3. Averaged numbers of iterations of (a) HT, (b) T1, (c) T2, (d) H, (e) DH in the missing data case, with varying sample size and SNR.

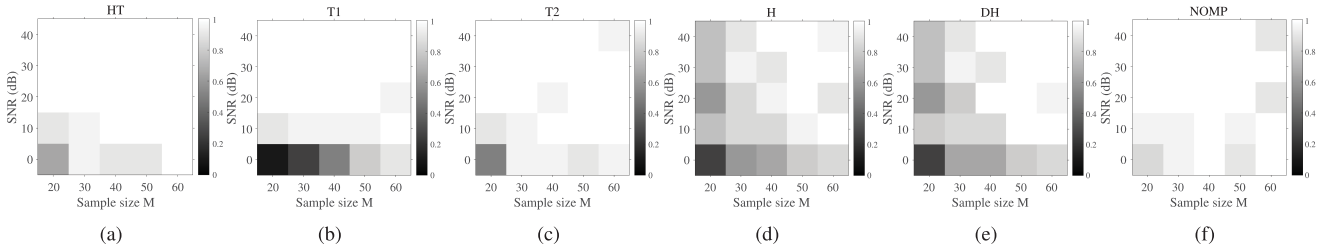


Fig. 4. Success rates of achieving global optimality of (a) HT, (b) T1, (c) T2, (d) H, (e) DH, (f) NOMP in the missing data case, with varying sample size and SNR.

The success rates of achieving global optimality are shown in Fig. 4. Once again, HT improves upon T1 and T2 and is comparable to NOMP. H and DH have degraded performance when the sample size is small. It can be shown that the failure of DH is typically caused by convergence to local minima rather than occurrence of symmetric damping factors, which is consistent with what is observed in [28].

Overall, the proposed HT approach has good performance with respect to both convergence and global optimality and realizes the MLE effectively in both full and missing data cases.

B. Statistical Efficiency

In this subsection, we present further numerical results to demonstrate the accuracy of the proposed HT approach. Besides the previous algorithms, we also compare with ESPRIT combined with forward-only and forward-backward spatial smoothing, denoted by FOSS-E and FBSS-E respectively, and the two CRBs mentioned in Section II-C. To compare with the CRBs, we fix the frequencies and coefficients in each experiment.

In *Experiment 3* we consider the full data case, with closely located frequencies and varying SNR. We fix the frequencies

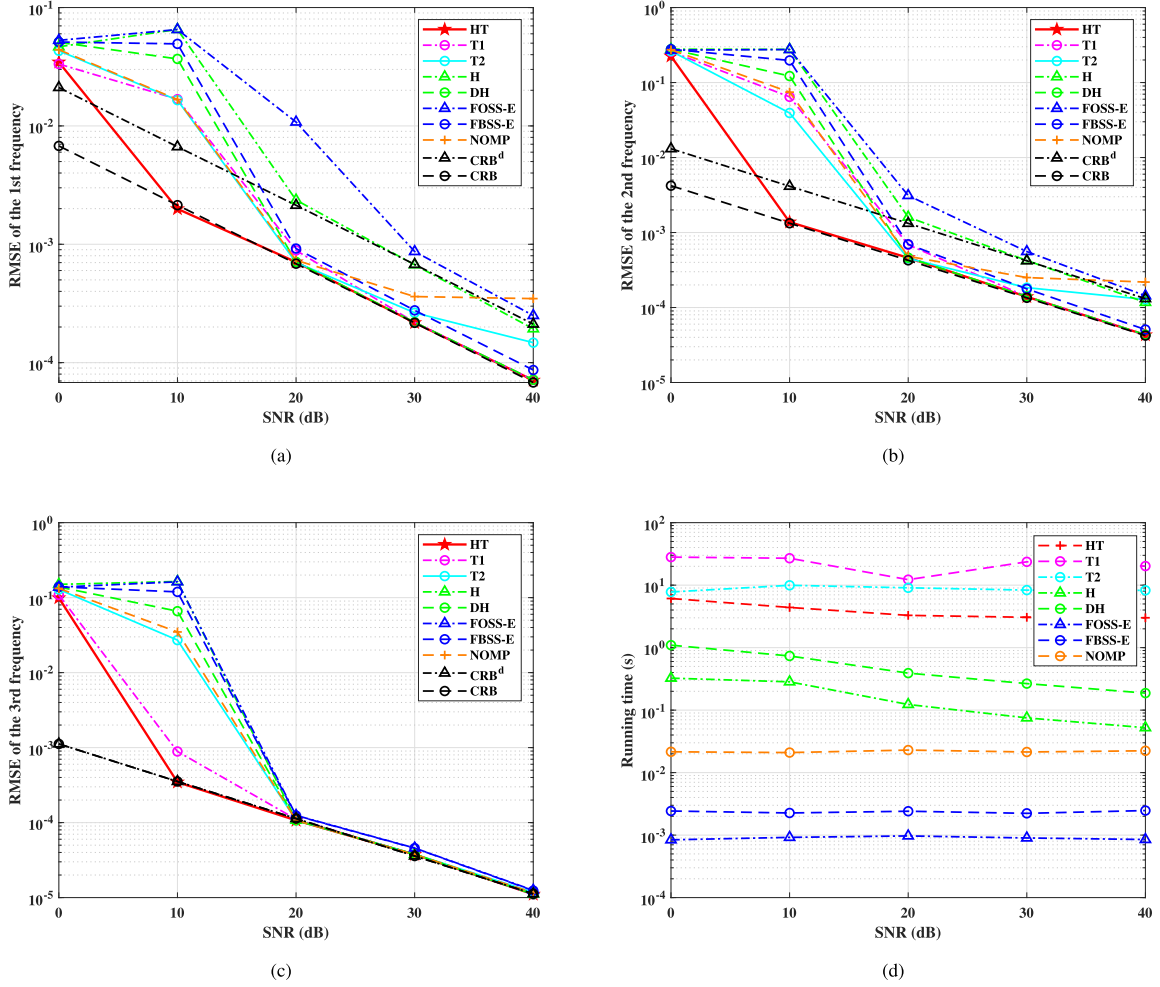


Fig. 5. RMSE of (a) 1st, (b) 2nd, and (c) 3rd frequency and (d) the running time of algorithms in the full data case.

$\mathbf{f}^* = [0.2, 0.2 + 0.5/N, 0.6]^T$ and coefficients $\mathbf{s}^* = [0.51 + 0.11i, -0.51 - 0.66i, -0.99 + 0.32i]^T$. A number of $N = 65$ noisy samples are acquired with varying SNR. The RMSE of frequency estimates is computed and averaged over 100 Monte Carlo runs for each SNR. In Fig. 5 we plot the RMSE curves together with square roots of the two CRBs for the frequencies. Note first that the true, undamped CRB is significantly lower than the damped CRB for the first two closely located frequencies, while they almost coincide for the third frequency. The proposed HT achieves the true CRB for SNR ≥ 10 dB, while that is possible for the other algorithms only when SNR ≥ 20 dB. Moreover, H and FOSS-E can only approach the damped CRB, and the DH and FBSS-E can be lower than the damped CRB by exploiting the knowledge of undampedness, which confirms our analysis in Section II-C. Note also that T2 and NOMP do not achieve the CRB in the high SNR regime for closely located frequencies; see also similar behaviors in Figs. 2 and 4. For T2, it is caused partly by convergence issues according to Figs. 1 and 3. For NOMP, such inaccuracy can be alleviated by choosing larger parameters γ, R_s, R_c at cost of more computations. We also plot the curves of CPU computational times in Fig. 5(d). It is seen that HT is faster than T1 and T2 but slower than the other approaches.

In *Experiment 4* we consider the missing data case with $N = 65$ and $M = 40$. We fix the frequencies $\mathbf{f}^* = [0.2, 0.2 + 0.5/N, 0.6]^T$ and coefficients $\mathbf{s}^* = [0.06 + 1.97i, 1.00 - 0.95i, -1.63 + 0.44i]^T$. The ESPRIT algorithm is not applicable in this case. In Fig. 6 we plot the error curves averaged over 100 Monte Carlo trials, where T1 and H are omitted due to poor performance. Similarly as in the full data case, HT achieves the CRB and has the best performance. The running time of HT gets close to DH in this case.

In *Experiment 5* we consider the estimation of 2-D frequencies given $M = 29$ noisy samples chosen randomly from $N_1 \times N_2 = 7 \times 7$ regularly spaced samples. The signal consists of $K = 2$ spectral components with $\mathbf{f}^* = \begin{bmatrix} 0.2 & 0.6 \\ 0.24 & 0.64 \end{bmatrix} \in \mathbb{R}^{K \times 2}$ and $\mathbf{s}^* = [1.33 + 1.07i, -2.68 - 0.24i]^T$. We plot in Fig. 7 the RMSE curves of both frequencies, computed using 100 Monte Carlo trials. The CRBs in this case are computed following [59]. The RMSE of H tends to the damped CRB, while that of DH is close to the undamped CRB. The RMSE of the proposed HT also approaches the undamped CRB and is better than that of T1 and T2.

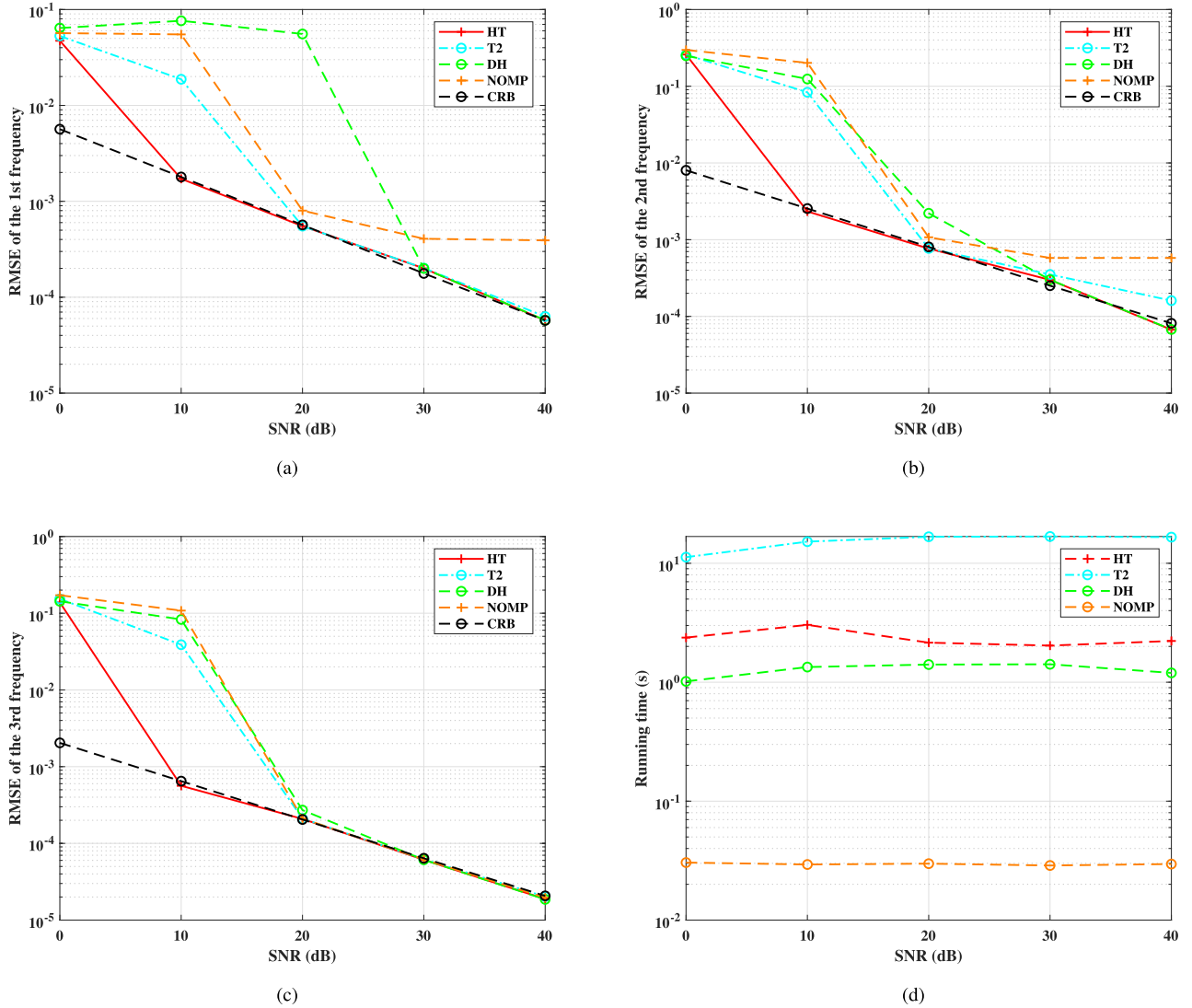


Fig. 6. RMSE of (a) 1st, (b) 2nd, and (c) 3rd frequency and (d) the running time of algorithms in the missing data case.

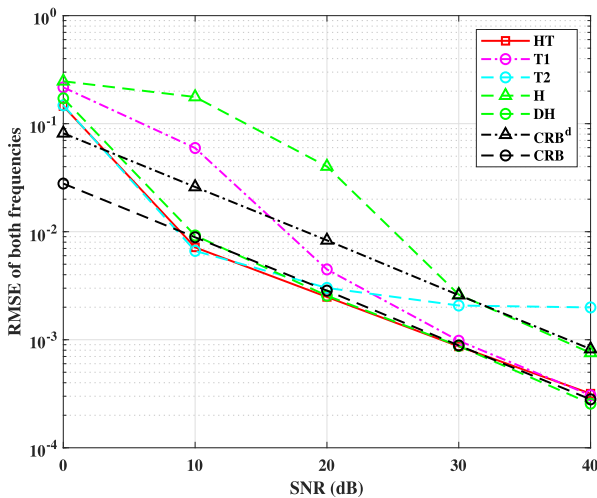


Fig. 7. RMSE of frequency estimates in the 2-D missing data case.

VII. CONCLUSION

In this paper, an exact maximum likelihood estimator for line spectral estimation in the signal domain was proposed based on rank-constrained structured matrix recovery and the ADMM algorithm. It is also usable in the missing data and arbitrary-dimensional cases. It was demonstrated via extensive numerical results that the proposed approach realizes the exact MLE and has good performance with respect to convergence, global optimality and statistical efficiency.

The proposed low-rank Hankel-Toeplitz method provides an exact and unique characterization of the spectral sparsity and can be used as a general technique in future spectral-sparsity-based approaches. The current ADMM algorithm needs to compute a truncated eigen-decomposition in each iteration. A future research direction will be to develop faster nonconvex algorithms based on the Hankel-Toeplitz method. Several references on nonconvex approaches to low-rank matrix recovery

with guaranteed global optimality [60], [61], [62] may prove useful in this direction.

APPENDIX

A. ADMM Updates

1) *ADMM for Toeplitz-1 (18)*: We make the substitutions

$$\mathbf{z} = \{\mathbf{x}, \mathbf{u}, \mathbf{t}\}, \quad h(\mathbf{z}) = \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \mathcal{M}\mathbf{z} = \begin{bmatrix} \mathbf{u} & \mathbf{x}^H \\ \mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix}.$$

The update for $\mathbf{Q}$ is given by:

$$\mathbf{Q}^{l+1} = \mathcal{P}_{\mathcal{S}_K} \left(\begin{bmatrix} \mathbf{u}^l & (\mathbf{x}^l)^H \\ \mathbf{x}^l & \mathcal{T}\mathbf{t}^l \end{bmatrix} - \mu^{-1} \mathbf{\Lambda}^l \right). \quad (61)$$

Equating the partial derivatives to zero yields the updates:

$$\mathbf{x}^{l+1} = (1 + \mu)^{-1} [\mathbf{y} + \mu (\mathbf{q}_2^{l+1} + \mu^{-1} \mathbf{\lambda}_2^l)], \quad (62)$$

$$\mathbf{u}^{l+1} = \mathbf{q}_1^{l+1} + \mu^{-1} \mathbf{\lambda}_1^l, \quad (63)$$

$$\mathbf{t}^{l+1} = \mathbf{D}_{\mathcal{T}}^{-1} \mathcal{T}^H (\mathbf{Q}_3^{l+1} + \mu^{-1} \mathbf{\Lambda}_3^l), \quad (64)$$

where $\mathbf{Q}^{l+1} = \begin{bmatrix} \mathbf{q}_1^{l+1} & (\mathbf{q}_2^{l+1})^H \\ \mathbf{q}_2^{l+1} & \mathbf{Q}_3^{l+1} \end{bmatrix}$, $\mathbf{\Lambda}^l = \begin{bmatrix} \mathbf{\lambda}_1^l & (\mathbf{\lambda}_2^l)^H \\ \mathbf{\lambda}_2^l & \mathbf{\Lambda}_3^l \end{bmatrix}$. In this case $\mathcal{T}^H$ and $\mathbf{D}_{\mathcal{T}}$ are defined as previously with the matrix size changed from $n \times n$ to $N \times N$. The update for $\mathbf{\Lambda}$ is given by:

$$\mathbf{\Lambda}^{l+1} = \mathbf{\Lambda}^l + \mu \left(\mathbf{Q}^{l+1} - \begin{bmatrix} \mathbf{u}^{l+1} & (\mathbf{x}^{l+1})^H \\ \mathbf{x}^{l+1} & \mathcal{T}\mathbf{t}^{l+1} \end{bmatrix} \right). \quad (65)$$

2) *ADMM for Toeplitz-2 (19)*: By making the substitutions

$$\mathbf{z} = \{\mathbf{x}, \mathbf{t}\}, \quad h(\mathbf{z}) = \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \mathcal{M}\mathbf{z} = \begin{bmatrix} \mathbf{t}_1 & \mathbf{x}^H \\ \mathbf{x} & \mathcal{T}\mathbf{t} \end{bmatrix},$$

we have

$$\mathbf{t}_1^{l+1} = (N + 1)^{-1} \text{tr} (\mathbf{Q}^{l+1} + \mu^{-1} \mathbf{\Lambda}^l), \quad (66)$$

$$\mathbf{t}_j^{l+1} = [\mathbf{D}_{\mathcal{T}}^{-1} \mathcal{T}^H (\mathbf{Q}_3^{l+1} + \mu^{-1} \mathbf{\Lambda}_3^l)]_j \quad (67)$$

for $j = 2, \dots, N$. The update for $\mathbf{x}$ remains unchanged, and $\mathbf{Q}, \mathbf{\Lambda}$ are updated similarly as (61) and (65) by changing $\mathbf{u}$ to $\mathbf{t}_1$.

3) *ADMM for Hankel (9)*: Letting

$$\mathbf{z} = \mathbf{x}, \quad h(\mathbf{z}) = \frac{1}{2} \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \mathcal{M}\mathbf{z} = \mathcal{H}\mathbf{x}, \quad (68)$$

we obtain the updates:

$$\mathbf{Q}^{l+1} = \mathcal{P}_{\mathcal{S}_K} (\mathcal{H}\mathbf{x}^l - \mu^{-1} \mathbf{\Lambda}^l), \quad (69)$$

$$\mathbf{x}^{l+1} = (\mathbf{I} + \mu \mathbf{D}_{\mathcal{H}})^{-1} [\mathbf{y} + \mu \mathcal{H}^H (\mathbf{Q}^{l+1} + \mu^{-1} \mathbf{\Lambda}^l)], \quad (70)$$

$$\mathbf{\Lambda}^{l+1} = \mathbf{\Lambda}^l + \mu (\mathbf{Q}^{l+1} - \mathcal{H}\mathbf{x}^{l+1}). \quad (71)$$

Note that in this case $\mathcal{S}_K$ is the set of matrices of rank no greater than K , and the projection onto $\mathcal{S}_K$ is obtained via the truncated SVD of the matrix argument by setting all but the largest K singular values to zero. The update rules in (70)–(71) have been derived in [26] and are presented here for completeness.

4) *ADMM for D-Hankel (11)*: In this case, we make the substitutions

$$\mathbf{z} = \mathbf{x}, \quad h(\mathbf{z}) = \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \mathcal{M}\mathbf{z} = [\mathcal{H}\mathbf{x} \mid \mathbf{J}\mathcal{H}\bar{\mathbf{x}}\mathbf{J}] \quad (72)$$

and obtain

$$\mathbf{Q}^{l+1} = \mathcal{P}_{\mathcal{S}_K} \left([\mathcal{H}\mathbf{x}^l \mid \mathbf{J}\mathcal{H}\bar{\mathbf{x}}^l\mathbf{J}] - \mu^{-1} \mathbf{\Lambda}^l \right),$$

$$\mathbf{x}^{l+1} = (\mathbf{I} + \mu \mathbf{D}_{\mathcal{H}})^{-1} \cdot$$

$$\left[\mathbf{y} + \frac{\mu}{2} \mathcal{H}^H (\mathbf{Q}_1^{l+1} + \mathbf{J}\bar{\mathbf{Q}}_2^{l+1}\mathbf{J} + \mu^{-1} (\mathbf{\Lambda}_1^l + \mathbf{J}\bar{\mathbf{\Lambda}}_2^l\mathbf{J})) \right],$$

$$\mathbf{\Lambda}^{l+1} = \mathbf{\Lambda}^l + \mu \left(\mathbf{Q}^{l+1} - [\mathcal{H}\mathbf{x}^{l+1} \mid \mathbf{J}\mathcal{H}\bar{\mathbf{x}}^{l+1}\mathbf{J}] \right),$$

where $\mathbf{Q}^{l+1} = [\mathbf{Q}_1^{l+1} \mid \mathbf{Q}_2^{l+1}]$, $\mathbf{\Lambda}^l = [\mathbf{\Lambda}_1^l \mid \mathbf{\Lambda}_2^l]$ and the projection $\mathcal{P}_{\mathcal{S}_K}$ is computed via the truncated SVD.

B. Proof of Theorem 2

By the optimality conditions, the sequences generated by (44)–(45) satisfy

$$\mathbf{0} \in \partial \delta_{\mathcal{S}_K} (\mathbf{Q}^{l+1}) + \mu (\mathbf{Q}^{l+1} - \mathcal{M}\mathbf{z}^l + \mu^{-1} \mathbf{\Lambda}^l),$$

$$\mathbf{0} = \nabla h(\mathbf{z}^{l+1}) - \mathcal{M}^H \mathbf{\Lambda}^l - \mu \mathcal{M}^H (\mathbf{Q}^{l+1} - \mathcal{M}\mathbf{z}^{l+1}). \quad (73)$$

Invoking the updates of $\mathbf{\Lambda}^{l+1}$ in (46), we have

$$-\mu (\mathcal{M}\mathbf{z}^{l+1} - \mathcal{M}\mathbf{z}^l) \in \partial \delta_{\mathcal{S}_K} (\mathbf{Q}^{l+1}) + \mathbf{\Lambda}^{l+1},$$

$$\nabla h(\mathbf{z}^{l+1}) = \mathcal{M}^H \mathbf{\Lambda}^{l+1},$$

$$\mathbf{Q}^{l+1} - \mathcal{M}\mathbf{z}^{l+1} = \mu^{-1} (\mathbf{\Lambda}^{l+1} - \mathbf{\Lambda}^l). \quad (74)$$

Suppose that $\{\mathbf{Q}^{l_j}, \mathbf{z}^{l_j}, \mathbf{\Lambda}^{l_j}\}$ is a subsequence that converges to the limit point $\{\mathbf{Q}^*, \mathbf{z}^*, \mathbf{\Lambda}^*\}$, i.e.,

$$\lim_{j \rightarrow \infty} \{\mathbf{Q}^{l_j}, \mathbf{z}^{l_j}, \mathbf{\Lambda}^{l_j}\} = \{\mathbf{Q}^*, \mathbf{z}^*, \mathbf{\Lambda}^*\}. \quad (75)$$

By the update of $\mathbf{Q}^{l_j}$ in (44), we have that $\mathbf{Q}^{l_j} \in \mathcal{S}_K$. Note that $\mathcal{S}_K$ is the intersection of two closed sets, the positive semidefinite cone and the set of matrices of rank no greater than K , and thus it is closed.¹ Consequently, $\mathbf{Q}^* \in \mathcal{S}_K$ and

$$\lim_{j \rightarrow \infty} \delta_{\mathcal{S}_K} (\mathbf{Q}^{l_j+1}) = \delta_{\mathcal{S}_K} (\mathbf{Q}^*) = 0. \quad (76)$$

Taking the limit in (74) along the subsequence and using (54) and (76), we obtain (55) and complete the proof.

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¹The proof of the closedness of the set of matrices of rank no greater than K is available online at <https://math.stackexchange.com/q/43194> or <https://math.stackexchange.com/q/1319547>.

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